

	$\mathcal{K}_{0^+}^{\#1}$	$\mathcal{K}_{0^-}^{\#1\alpha\beta\chi}$				
$\mathcal{K}_{0^+}^{\#1\dagger}$	0	0				
$\mathcal{K}_{0^-}^{\#1\dagger\alpha\beta\chi}$	0	0	$\mathcal{K}_{1^+}^{\#1\alpha\beta}$	$\mathcal{K}_{1^+}^{\#2\alpha\beta}$	$\mathcal{K}_{1^-}^{\#1\alpha}$	$\mathcal{K}_{1^-}^{\#2\alpha}$
$\mathcal{K}_{1^+}^{\#1\dagger\alpha\beta}$	$\Upsilon_1 k^2$	$-\sqrt{2}\Upsilon_1 k^2$	0	0		
$\mathcal{K}_{1^+}^{\#2\dagger\alpha\beta}$	$-\sqrt{2}\Upsilon_1 k^2$	$2\Upsilon_1 k^2$	0	0		
$\mathcal{K}_{1^-}^{\#1\dagger\alpha}$	0	0	$\frac{\Upsilon_2 k^2}{2}$	$-\frac{\Upsilon_3 k^2}{2\sqrt{2}}$		
$\mathcal{K}_{1^-}^{\#2\dagger\alpha}$	0	0	$-\frac{\Upsilon_3 k^2}{2\sqrt{2}}$	$\frac{\Upsilon_4 k^2}{2}$	$\mathcal{K}_{2^+}^{\#1\alpha\beta}$	$\mathcal{K}_{2^-}^{\#1\alpha\beta\chi}$
			$\mathcal{K}_{2^+}^{\#1\dagger\alpha\beta}$	0	0	
			$\mathcal{K}_{2^-}^{\#1\dagger\alpha\beta\chi}$	0	$\frac{3k^2\langle K_{15}^{(4)}\rangle}{2}$	

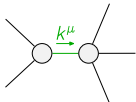
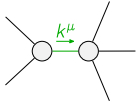
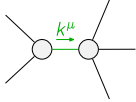
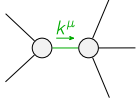
	$\mathcal{J}_{0^+}^{\#1}$	$\mathcal{J}_{0^-}^{\#1\alpha\beta\chi}$				
$\mathcal{J}_{0^+}^{\#1\dagger}$	0	0				
$\mathcal{J}_{0^-}^{\#1\dagger\alpha\beta\chi}$	0	0	$\mathcal{J}_{1^+}^{\#1\alpha\beta}$	$\mathcal{J}_{1^+}^{\#2\alpha\beta}$	$\mathcal{J}_{1^-}^{\#1\alpha}$	$\mathcal{J}_{1^-}^{\#2\alpha}$
$\mathcal{J}_{1^+}^{\#1\dagger\alpha\beta}$	$\frac{1}{3\text{Det}(1^+)}$	$-\frac{\sqrt{2}}{3\text{Det}(1^+)}$	0	0		
$\mathcal{J}_{1^+}^{\#2\dagger\alpha\beta}$	$-\frac{\sqrt{2}}{3\text{Det}(1^+)}$	$\frac{2}{3\text{Det}(1^+)}$	0	0		
$\mathcal{J}_{1^-}^{\#1\dagger\alpha}$	0	0	$\frac{\Upsilon_4 k^2}{2\text{Det}(1^-)}$	$\frac{\Upsilon_3 k^2}{2\sqrt{2}\text{Det}(1^-)}$		
$\mathcal{J}_{1^-}^{\#2\dagger\alpha}$	0	0	$\frac{\Upsilon_3 k^2}{2\sqrt{2}\text{Det}(1^-)}$	$\frac{\Upsilon_2 k^2}{2\text{Det}(1^-)}$	$\mathcal{J}_{2^+}^{\#1\alpha\beta}$	$\mathcal{J}_{2^-}^{\#1\alpha\beta\chi}$
			$\mathcal{J}_{2^+}^{\#1\dagger\alpha\beta}$	0	0	
			$\mathcal{J}_{2^-}^{\#1\dagger\alpha\beta\chi}$	0	$\frac{2}{3k^2\langle K_{15}^{(4)}\rangle}$	

Abbreviations used in matrices

$$\begin{aligned} \Upsilon_1 &== 2\langle K_{15}^{(4)}\rangle - \langle K_3^{(4)}\rangle \&\& \Upsilon_2 == 2\langle K_1^{(4)}\rangle + 3\langle K_{15}^{(4)}\rangle \&\& \Upsilon_3 == 2\langle K_1^{(4)}\rangle - \langle K_7^{(4)}\rangle \&\& \\ \Upsilon_4 &== \langle K_1^{(4)}\rangle + 15\langle K_{15}^{(4)}\rangle - 9\langle K_3^{(4)}\rangle - \langle K_7^{(4)}\rangle \&\& \text{Det}(1^+) == 3k^2(2\langle K_{15}^{(4)}\rangle - \langle K_3^{(4)}\rangle) \end{aligned}$$

Lagrangian

$$\begin{aligned} &\langle K_1^{(4)}\rangle\partial_\beta\mathcal{K}_{\chi\delta}^{\delta}\partial^{\chi}\mathcal{K}_{\alpha}^{\alpha\beta}-\langle K_1^{(4)}\rangle\partial_{\chi}\mathcal{K}_{\beta\delta}^{\delta}\partial^{\chi}\mathcal{K}_{\alpha}^{\alpha\beta}+\langle K_3^{(4)}\rangle\partial_{\beta}\mathcal{K}^{\alpha\beta\chi}\partial_{\delta}\mathcal{K}_{\alpha\chi}^{\delta}-6\langle K_{15}^{(4)}\rangle\partial_{\alpha}\mathcal{K}^{\alpha\beta\chi}\partial_{\delta}\mathcal{K}_{\beta\chi}^{\delta}+ \\ &4\langle K_3^{(4)}\rangle\partial_{\alpha}\mathcal{K}^{\alpha\beta\chi}\partial_{\delta}\mathcal{K}_{\beta\chi}^{\delta}+3\langle K_{15}^{(4)}\rangle\partial_{\beta}\mathcal{K}^{\alpha\beta\chi}\partial_{\delta}\mathcal{K}_{\chi\alpha}^{\delta}-\langle K_3^{(4)}\rangle\partial_{\beta}\mathcal{K}^{\alpha\beta\chi}\partial_{\delta}\mathcal{K}_{\chi\alpha}^{\delta}+\langle K_7^{(4)}\rangle\partial^{\chi}\mathcal{K}_{\alpha}^{\alpha\beta}\partial_{\delta}\mathcal{K}_{\chi\beta}^{\delta}+ \\ &3\langle K_{15}^{(4)}\rangle\partial_{\alpha}\mathcal{K}^{\alpha\beta\chi}\partial_{\delta}\mathcal{K}_{\beta\chi}^{\delta}-2\langle K_3^{(4)}\rangle\partial_{\alpha}\mathcal{K}^{\alpha\beta\chi}\partial_{\delta}\mathcal{K}_{\beta\chi}^{\delta}+\langle K_{15}^{(4)}\rangle\partial_{\delta}\mathcal{K}_{\alpha\beta\chi}\partial^{\delta}\mathcal{K}^{\alpha\beta\chi}+\langle K_{15}^{(4)}\rangle\partial_{\delta}\mathcal{K}_{\beta\alpha\chi}\partial^{\delta}\mathcal{K}^{\alpha\beta\chi} \end{aligned}$$

Added source term(s):	$\mathcal{K}^{\alpha\beta\chi}\mathcal{J}_{\alpha\beta\chi}$		
Source constraint(s)	# constraint(s)	Covariant form	
$\mathcal{J}_{0^-}^{\#1\alpha\beta\chi} == 0$	1	$\partial_{\delta}\partial^{\alpha}\mathcal{J}^{\beta\chi\delta}+\partial_{\delta}\partial^{\alpha}\mathcal{J}^{\delta\beta\chi}+\partial_{\delta}\partial^{\beta}\mathcal{J}^{\chi\alpha\delta}+\partial_{\delta}\partial^{\chi}\mathcal{J}^{\alpha\beta\delta}+\partial_{\delta}\partial^{\chi}\mathcal{J}^{\delta\alpha\beta}+\partial_{\delta}\partial^{\delta}\mathcal{J}^{\beta\alpha\chi}==$ $\partial_{\delta}\partial^{\alpha}\mathcal{J}^{\chi\beta\delta}+\partial_{\delta}\partial^{\beta}\mathcal{J}^{\alpha\chi\delta}+\partial_{\delta}\partial^{\beta}\mathcal{J}^{\delta\alpha\chi}+\partial_{\delta}\partial^{\chi}\mathcal{J}^{\beta\alpha\delta}+\partial_{\delta}\partial^{\delta}\mathcal{J}^{\alpha\beta\chi}+\partial_{\delta}\partial^{\delta}\mathcal{J}^{\chi\alpha\beta}$	
$\mathcal{J}_{0^+}^{\#1} == 0$	1	$\partial_{\beta}\mathcal{J}_{\alpha}^{\alpha\beta} == 0$	
$2\mathcal{J}_{1^+}^{\#1\alpha\beta}+\mathcal{J}_{1^+}^{\#2\alpha\beta} == 0$	3	$\partial_{\chi}\mathcal{J}^{\alpha\beta\chi}+\partial_{\chi}\mathcal{J}^{\chi\alpha\beta} == \partial_{\chi}\mathcal{J}^{\beta\alpha\chi}$	
$\mathcal{J}_{2^+}^{\#1\alpha\beta} == 0$	5	$3\partial_{\delta}\partial_{\chi}\partial^{\alpha}\mathcal{J}^{\chi\beta\delta}+3\partial_{\delta}\partial_{\chi}\partial^{\beta}\mathcal{J}^{\chi\alpha\delta}+2\eta^{\alpha\beta}\partial_{\epsilon}\partial^{\epsilon}\partial_{\delta}\mathcal{J}^{\chi\chi\delta}==2\partial_{\delta}\partial^{\beta}\partial^{\alpha}\mathcal{J}^{\chi\chi\delta}+3(\partial_{\delta}\partial^{\delta}\partial_{\chi}\mathcal{J}^{\alpha\beta\chi}+\partial_{\delta}\partial^{\delta}\partial_{\chi}\mathcal{J}^{\beta\alpha\chi})$	
Total # constraint(s):	10		
Resolved pole(s)	# polarization(s)	Square mass	Residue
	1	0	$(528\langle K_1^{(4)}\rangle\langle K_{15}^{(4)}\rangle^2+720\langle K_{15}^{(4)}\rangle^3-552\langle K_1^{(4)}\rangle\langle K_{15}^{(4)}\rangle\langle K_3^{(4)}\rangle-792\langle K_{15}^{(4)}\rangle^2\langle K_3^{(4)}\rangle+144\langle K_1^{(4)}\rangle\langle K_3^{(4)}\rangle^2+216\langle K_{15}^{(4)}\rangle\langle K_3^{(4)}\rangle^2-48\langle K_{15}^{(4)}\rangle^2\langle K_7^{(4)}\rangle+24\langle K_{15}^{(4)}\rangle\langle K_3^{(4)}\rangle\langle K_7^{(4)}\rangle-8\langle K_{15}^{(4)}\rangle\langle K_7^{(4)}\rangle^2+4\langle K_3^{(4)}\rangle\langle K_7^{(4)}\rangle^2-\sqrt{((105\langle K_{15}^{(4)}\rangle^2-108\langle K_{15}^{(4)}\rangle\langle K_3^{(4)}\rangle+28\langle K_3^{(4)}\rangle^2)(-66\langle K_1^{(4)}\rangle\langle K_{15}^{(4)}\rangle-90\langle K_{15}^{(4)}\rangle^2+36\langle K_1^{(4)}\rangle\langle K_3^{(4)}\rangle+54\langle K_{15}^{(4)}\rangle\langle K_3^{(4)}\rangle+6\langle K_{15}^{(4)}\rangle\langle K_7^{(4)}\rangle+\langle K_7^{(4)}\rangle^2)})/(K_{15}^{(4)}(2\langle K_{15}^{(4)}\rangle-\langle K_3^{(4)}\rangle)(90\langle K_{15}^{(4)}\rangle^2+6\langle K_1^{(4)}\rangle(11\langle K_{15}^{(4)}\rangle-6\langle K_3^{(4)}\rangle)-\langle K_7^{(4)}\rangle^2-6\langle K_{15}^{(4)}\rangle(9\langle K_3^{(4)}\rangle+\langle K_7^{(4)}\rangle)))$
	1	0	$(528\langle K_1^{(4)}\rangle\langle K_{15}^{(4)}\rangle^2+720\langle K_{15}^{(4)}\rangle^3-552\langle K_1^{(4)}\rangle\langle K_{15}^{(4)}\rangle\langle K_3^{(4)}\rangle-792\langle K_{15}^{(4)}\rangle^2\langle K_3^{(4)}\rangle+144\langle K_1^{(4)}\rangle\langle K_3^{(4)}\rangle^2+216\langle K_{15}^{(4)}\rangle\langle K_3^{(4)}\rangle^2-48\langle K_{15}^{(4)}\rangle^2\langle K_7^{(4)}\rangle+24\langle K_{15}^{(4)}\rangle\langle K_3^{(4)}\rangle\langle K_7^{(4)}\rangle-8\langle K_{15}^{(4)}\rangle\langle K_7^{(4)}\rangle^2+4\langle K_3^{(4)}\rangle\langle K_7^{(4)}\rangle^2+\sqrt{((105\langle K_{15}^{(4)}\rangle^2-108\langle K_{15}^{(4)}\rangle\langle K_3^{(4)}\rangle+28\langle K_3^{(4)}\rangle^2)(-66\langle K_1^{(4)}\rangle\langle K_{15}^{(4)}\rangle-90\langle K_{15}^{(4)}\rangle^2+36\langle K_1^{(4)}\rangle\langle K_3^{(4)}\rangle+54\langle K_{15}^{(4)}\rangle\langle K_3^{(4)}\rangle+6\langle K_{15}^{(4)}\rangle\langle K_7^{(4)}\rangle+\langle K_7^{(4)}\rangle^2)})/(K_{15}^{(4)}(2\langle K_{15}^{(4)}\rangle-\langle K_3^{(4)}\rangle)(90\langle K_{15}^{(4)}\rangle^2+6\langle K_1^{(4)}\rangle(11\langle K_{15}^{(4)}\rangle-6\langle K_3^{(4)}\rangle)-\langle K_7^{(4)}\rangle^2-6\langle K_{15}^{(4)}\rangle(9\langle K_3^{(4)}\rangle+\langle K_7^{(4)}\rangle)))$
	2	0	$-((756\langle K_1^{(4)}\rangle\langle K_{15}^{(4)}\rangle^2+2196\langle K_{15}^{(4)}\rangle^3-864\langle K_1^{(4)}\rangle\langle K_{15}^{(4)}\rangle\langle K_3^{(4)}\rangle-2556\langle K_{15}^{(4)}\rangle^2\langle K_3^{(4)}\rangle+252\langle K_1^{(4)}\rangle\langle K_3^{(4)}\rangle^2+756\langle K_{15}^{(4)}\rangle\langle K_3^{(4)}\rangle^2-132\langle K_{15}^{(4)}\rangle^2\langle K_7^{(4)}\rangle+84\langle K_{15}^{(4)}\rangle\langle K_3^{(4)}\rangle\langle K_7^{(4)}\rangle-8\langle K_{15}^{(4)}\rangle\langle K_7^{(4)}\rangle^2+7\langle K_3^{(4)}\rangle\langle K_7^{(4)}\rangle^2+\sqrt{(2178576\langle K_{15}^{(4)}\rangle^6+1296\langle K_1^{(4)}\rangle^2(21\langle K_{15}^{(4)}\rangle^2-24\langle K_{15}^{(4)}\rangle\langle K_3^{(4)}\rangle+7\langle K_3^{(4)}\rangle^2)^2+49\langle K_3^{(4)}\rangle^2\langle K_7^{(4)}\rangle^4-247968\langle K_{15}^{(4)}\rangle^5(21\langle K_3^{(4)}\rangle+\langle K_7^{(4)}\rangle)+432\langle K_{15}^{(4)}\rangle^4(10893\langle K_3^{(4)}\rangle+1096\langle K_3^{(4)}\rangle\langle K_7^{(4)}\rangle-27\langle K_7^{(4)}\rangle^2)+8\langle K_{15}^{(4)}\rangle\langle K_3^{(4)}\rangle\langle K_7^{(4)}\rangle^2(675\langle K_3^{(4)}\rangle^2+75\langle K_3^{(4)}\rangle\langle K_7^{(4)}\rangle-23\langle K_7^{(4)}\rangle^2)-24\langle K_{15}^{(4)}\rangle^3(79380\langle K_3^{(4)}\rangle^3+12600\langle K_3^{(4)}\rangle^2\langle K_7^{(4)}\rangle-1107\langle K_3^{(4)}\rangle\langle K_7^{(4)}\rangle^2-52\langle K_7^{(4)}\rangle^3)+8\langle K_{15}^{(4)}\rangle^2(36450\langle K_3^{(4)}\rangle^4+8100\langle K_3^{(4)}\rangle^3\langle K_7^{(4)}\rangle-2565\langle K_3^{(4)}\rangle^2\langle K_7^{(4)}\rangle^2-201\langle K_3^{(4)}\rangle\langle K_7^{(4)}\rangle^3+26\langle K_7^{(4)}\rangle^4)+72\langle K_1^{(4)}\rangle(19188\langle K_{15}^{(4)}\rangle^5+49\langle K_3^{(4)}\rangle^3\langle K_7^{(4)}\rangle^2-108\langle K_{15}^{(4)}\rangle^4(431\langle K_3^{(4)}\rangle+11\langle K_7^{(4)}\rangle)-9\langle K_{15}^{(4)}\rangle^2\langle K_3^{(4)}\rangle(1940\langle K_3^{(4)}\rangle^2+156\langle K_3^{(4)}\rangle\langle K_7^{(4)}\rangle-53\langle K_7^{(4)}\rangle^2)+12\langle K_{15}^{(4)}\rangle^3(3552\langle K_3^{(4)}\rangle^2+185\langle K_3^{(4)}\rangle\langle K_7^{(4)}\rangle-25\langle K_7^{(4)}\rangle^2)+20\langle K_{15}^{(4)}\rangle\langle K_3^{(4)}\rangle^2(135\langle K_3^{(4)}\rangle^2+15\langle K_3^{(4)}\rangle\langle K_7^{(4)}\rangle-13\langle K_7^{(4)}\rangle^2))))/(K_{15}^{(4)}(2\langle K_{15}^{(4)}\rangle-\langle K_3^{(4)}\rangle)(90\langle K_{15}^{(4)}\rangle^2+6\langle K_1^{(4)}\rangle(11\langle K_{15}^{(4)}\rangle-6\langle K_3^{(4)}\rangle)-\langle K_7^{(4)}\rangle^2-6\langle K_{15}^{(4)}\rangle(9\langle K_3^{(4)}\rangle+\langle K_7^{(4)}\rangle)))$
	2	0	$(-756\langle K_1^{(4)}\rangle\langle K_{15}^{(4)}\rangle^2-2196\langle K_{15}^{(4)}\rangle^3+864\langle K_1^{(4)}\rangle\langle K_{15}^{(4)}\rangle\langle K_3^{(4)}\rangle+2556\langle K_{15}^{(4)}\rangle^2\langle K_3^{(4)}\rangle-252\langle K_1^{(4)}\rangle\langle K_3^{(4)}\rangle^2-756\langle K_{15}^{(4)}\rangle\langle K_3^{(4)}\rangle^2+132\langle K_{15}^{(4)}\rangle^2\langle K_7^{(4)}\rangle-84\langle K_{15}^{(4)}\rangle\langle K_3^{(4)}\rangle\langle K_7^{(4)}\rangle+8\langle K_{15}^{(4)}\rangle\langle K_7^{(4)}\rangle^2-7\langle K_3^{(4)}\rangle\langle K_7^{(4)}\rangle^2+\sqrt{(2178576\langle K_{15}^{(4)}\rangle^6+1296\langle K_1^{(4)}\rangle^2(21\langle K_{15}^{(4)}\rangle^2-24\langle K_{15}^{(4)}\rangle\langle K_3^{(4)}\rangle+7\langle K_3^{(4)}\rangle^2)^2+49\langle K_3^{(4)}\rangle^2\langle K_7^{(4)}\rangle^4-247968\langle K_{15}^{(4)}\rangle^5(21\langle K_3^{(4)}\rangle+\langle K_7^{(4)}\rangle)+432\langle K_{15}^{(4)}\rangle^4(10893\langle K_3^{(4)}\rangle+1096\langle K_3^{(4)}\rangle\langle K_7^{(4)}\rangle-27\langle K_7^{(4)}\rangle^2)+8\langle K_{15}^{(4)}\rangle\langle K_3^{(4)}\rangle\langle K_7^{(4)}\rangle^2(675\langle K_3^{(4)}\rangle^2+75\langle K_3^{(4)}\rangle\langle K_7^{(4)}\rangle-23\langle K_7^{(4)}\rangle^2)-24\langle K_{15}^{(4)}\rangle^3(79380\langle K_3^{(4)}\rangle^3+12600\langle K_3^{(4)}\rangle^2\langle K_7^{(4)}\rangle-1107\langle K_3^{(4)}\rangle\langle K_7^{(4)}\rangle^2-52\langle K_7^{(4)}\rangle^3)+8\langle K_{15}^{(4)}\rangle^2(36450\langle K_3^{(4)}\rangle^4+8100\langle K_3^{(4)}\rangle^3\langle K_7^{(4)}\rangle-2565\langle K_3^{(4)}\rangle^2\langle K_7^{(4)}\rangle^2-201\langle K_3^{(4)}\rangle\langle K_7^{(4)}\rangle^3+26\langle K_7^{(4)}\rangle^4)+72\langle K_1^{(4)}\rangle(19188\langle K_{15}^{(4)}\rangle^5+49\langle K_3^{(4)}\rangle^3\langle K_7^{(4)}\rangle^2-108\langle K_{15}^{(4)}\rangle^4(431\langle K_3^{(4)}\rangle+11\langle K_7^{(4)}\rangle)-9\langle K_{15}^{(4)}\rangle^2\langle K_3^{(4)}\rangle(1940\langle K_3^{(4)}\rangle^2+156\langle K_3^{(4)}\rangle\langle K_7^{(4)}\rangle-53\langle K_7^{(4)}\rangle^2)+12\langle K_{15}^{(4)}\rangle^3(3552\langle K_3^{(4)}\rangle^2+185\langle K_3^{(4)}\rangle\langle K_7^{(4)}\rangle-25\langle K_7^{(4)}\rangle^2)+20\langle K_{15}^{(4)}\rangle\langle K_3^{(4)}\rangle^2(135\langle K_3^{(4)}\rangle^2+15\langle K_3^{(4)}\rangle\langle K_7^{(4)}\rangle-13\langle K_7^{(4)}\rangle^2))))/(K_{15}^{(4)}(2\langle K_{15}^{(4)}\rangle-\langle K_3^{(4)}\rangle)(90\langle K_{15}^{(4)}\rangle^2+6\langle K_1^{(4)}\rangle(11\langle K_{15}^{(4)}\rangle-6\langle K_3^{(4)}\rangle)-\langle K_7^{(4)}\rangle^2-6\langle K_{15}^{(4)}\rangle(9\langle K_3^{(4)}\rangle+\langle K_7^{(4)}\rangle)))$
Resolved unitarity condition(s):	(Demonstrably impossible)		